Expelexia Lab - Data Analysis Report

File Name: medical_patient_data.csv

Date of Analysis: 2026-03-24 11:01

Prepared For: Expelexia (AI Analytics Program)

Project: Expelexia Lab

This report provides a comprehensive analysis and AI-generated insights based on the uploaded data.

Executive Summary

This report summarizes analysis results for medical_patient_data.csv using the same data shown in the website dashboard.
The dataset contains 944 records across 10 variables.
Main areas needing attention: footfall: 18 anomalies, high variance, tempMode: high variance, AQ: high variance.
Observed trend signals: footfall down, tempMode up, AQ down.
Recommendations focus on practical monitoring, early detection of unusual changes, and improving reliability of future data collection.

Data Overview

Measure	Value	What it means
Number of records	944	How many entries were analyzed.
Number of variables	10	How many fields were evaluated.
Missing values	{'footfall': 0, 'tempMode': 0, 'AQ': 0, 'USS': 0, 'CS': 0, 'VOC': 0, 'RP': 0, 'IP': 0, 'Temperature': 0, 'fail': 0}	Missing entries can reduce result reliability if frequent.

Website Dashboard Alignment

This table mirrors key trend, focus, and confidence indicators used in the website dashboard view.

Metric	Trend	Focus Area	Confidence
footfall	down	18 anomalies, high variance	0.980
tempMode	up	high variance	1.000
AQ	down	high variance	1.000
USS	up	high variance	1.000
CS	down	13 anomalies, high variance	0.990
VOC	up	high variance	1.000
RP	up	high variance	1.000

Metric	Trend	Focus Area	Confidence
IP	down	high variance	1.000
Temperature	up	high variance	1.000
fail	stable	high variance	1.000

Visual Analysis

The chart below shows how key values compare. It helps identify increases, decreases, and unusual patterns over time.

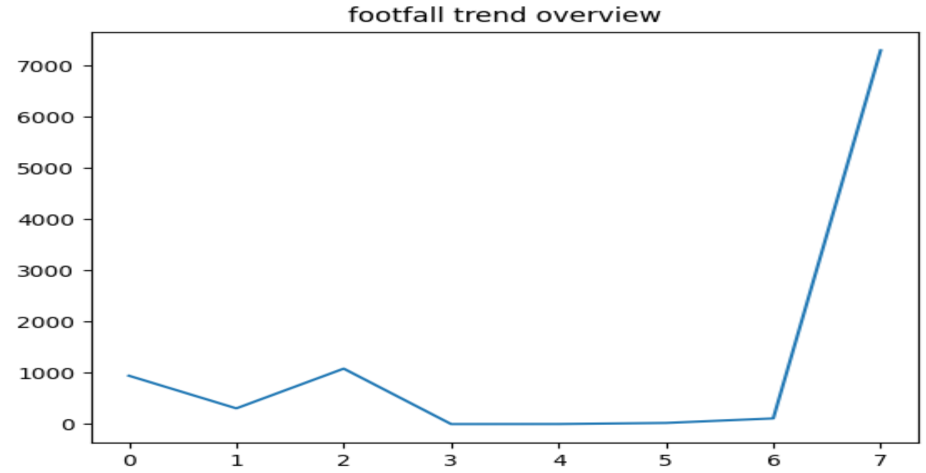

Chart Interpretation:
- footfall: bar chart centers near mean 306.381; line trend indicates down. Mean/median relation suggests a right-skewed distribution.
- tempMode: bar chart centers near mean 3.728; line trend indicates up. Mean/median relation suggests a right-skewed distribution.
- AQ: bar chart centers near mean 4.325; line trend indicates down. Mean/median relation suggests a right-skewed distribution.
- USS: bar chart centers near mean 2.940; line trend indicates up. Mean/median relation suggests a left-skewed distribution.
- CS: bar chart centers near mean 5.394; line trend indicates down. Mean/median relation suggests a left-skewed distribution.
- VOC: bar chart centers near mean 2.842; line trend indicates up. Mean/median relation suggests a right-skewed distribution.
- RP: bar chart centers near mean 47.043; line trend indicates up. Mean/median relation suggests a right-skewed distribution.
- IP: bar chart centers near mean 4.566; line trend indicates down. Mean/median relation suggests a right-skewed distribution.
- Temperature: bar chart centers near mean 16.332; line trend indicates up. Mean/median relation suggests a left-skewed distribution.
- fail: bar chart centers near mean 0.416; line trend indicates stable. Mean/median relation suggests a right-skewed distribution.
- Histogram bins show where value density is concentrated between quartile ranges.
- Pie chart summarizes stable vs attention-needed metrics from focus-area rules.

Data Change Explanation

Data Change Analysis:
- footfall: value range 0.000 to 7300.000 (spread 7300.000), middle 50% spans 109.000. Trend is down; volatility appears high; focus area is 18 anomalies, high variance.
- tempMode: value range 0.000 to 7.000 (spread 7.000), middle 50% spans 6.000. Trend is up; volatility appears high; focus area is high variance.
- AQ: value range 1.000 to 7.000 (spread 6.000), middle 50% spans 3.000. Trend is down; volatility appears high; focus area is high variance.
- USS: value range 1.000 to 7.000 (spread 6.000), middle 50% spans 2.000. Trend is up; volatility appears high; focus area is high variance.
- CS: value range 1.000 to 7.000 (spread 6.000), middle 50% spans 1.000. Trend is down; volatility appears high; focus area is 13 anomalies, high variance.
- VOC: value range 0.000 to 6.000 (spread 6.000), middle 50% spans 4.000. Trend is up; volatility appears high; focus area is high variance.
- RP: value range 19.000 to 91.000 (spread 72.000), middle 50% spans 24.000. Trend is up; volatility appears high; focus area is high variance.
- IP: value range 1.000 to 7.000 (spread 6.000), middle 50% spans 3.000. Trend is down; volatility appears high; focus area is high variance.
- Temperature: value range 1.000 to 24.000 (spread 23.000), middle 50% spans 7.000. Trend is up; volatility appears high; focus area is high variance.
- fail: value range 0.000 to 1.000 (spread 1.000), middle 50% spans 1.000. Trend is stable; volatility appears high; focus area is high variance.

Key Findings

• footfall: values range from 0.00 to 7300.00, average around 306.38. Trend appears down; review note: 18 anomalies, high variance.

• tempMode: values range from 0.00 to 7.00, average around 3.73. Trend appears up; review note: high variance.

• AQ: values range from 1.00 to 7.00, average around 4.33. Trend appears down; review note: high variance.

• USS: values range from 1.00 to 7.00, average around 2.94. Trend appears up; review note: high variance.

• CS: values range from 1.00 to 7.00, average around 5.39. Trend appears down; review note: 13 anomalies, high variance.

• VOC: values range from 0.00 to 6.00, average around 2.84. Trend appears up; review note: high variance.

AI-Powered Insights

Executive Summary: This is a plain-language summary.

Data Overview: This section explains the data in simple terms.

Technical Review (Advanced)

Technical Review (Advanced):
- footfall: mean=306.3814, std=1082.6067, min=0.0000, max=7300.0000, IQR=109.0000, CV=3.5335, trend=down, focus=18 anomalies, high variance, confidence=0.98.
- tempMode: mean=3.7278, std=2.6772, min=0.0000, max=7.0000, IQR=6.0000, CV=0.7182, trend=up, focus=high variance, confidence=1.0.

- AQ: mean=4.3252, std=1.4384, min=1.0000, max=7.0000, IQR=3.0000, CV=0.3326, trend=down, focus=high variance, confidence=1.0.
- USS: mean=2.9396, std=1.3837, min=1.0000, max=7.0000, IQR=2.0000, CV=0.4707, trend=up, focus=high variance, confidence=1.0.
- CS: mean=5.3941, std=1.2693, min=1.0000, max=7.0000, IQR=1.0000, CV=0.2353, trend=down, focus=13 anomalies, high variance, confidence=0.99.
- VOC: mean=2.8422, std=2.2733, min=0.0000, max=6.0000, IQR=4.0000, CV=0.7999, trend=up, focus=high variance, confidence=1.0.
- RP: mean=47.0434, std=16.4231, min=19.0000, max=91.0000, IQR=24.0000, CV=0.3491, trend=up, focus=high variance, confidence=1.0.
- IP: mean=4.5657, std=1.5993, min=1.0000, max=7.0000, IQR=3.0000, CV=0.3503, trend=down, focus=high variance, confidence=1.0.
- Temperature: mean=16.3316, std=5.9748, min=1.0000, max=24.0000, IQR=7.0000, CV=0.3658, trend=up, focus=high variance, confidence=1.0.
- fail: mean=0.4163, std=0.4932, min=0.0000, max=1.0000, IQR=1.0000, CV=1.1847, trend=stable, focus=high variance, confidence=1.0.

Technical Recommendations (Advanced)

Technical Recommendations (Advanced):
- footfall: high coefficient of variation (CV=3.5335); apply robust thresholding and recalibration checks.
- footfall: anomaly-heavy profile detected; run outlier root-cause analysis against raw capture logs.
- footfall: persistent downward trend; validate process degradation assumptions and lower tolerance thresholds.
- tempMode: high coefficient of variation (CV=0.7182); apply robust thresholding and recalibration checks.
- tempMode: persistent upward trend; evaluate drift alarms and upper tolerance boundaries.
- AQ: high coefficient of variation (CV=0.3326); apply robust thresholding and recalibration checks.
- AQ: persistent downward trend; validate process degradation assumptions and lower tolerance thresholds.
- USS: high coefficient of variation (CV=0.4707); apply robust thresholding and recalibration checks.
- USS: persistent upward trend; evaluate drift alarms and upper tolerance boundaries.
- CS: anomaly-heavy profile detected; run outlier root-cause analysis against raw capture logs.
- CS: persistent downward trend; validate process degradation assumptions and lower tolerance thresholds.
- VOC: high coefficient of variation (CV=0.7999); apply robust thresholding and recalibration checks.

Special AI Recommendations

Priority actions below are generated from trend volatility, confidence, and focus-area signals.

Abstract: The analysis processed 944 rows across 10 columns to evaluate quality, trend behavior, and operational risk.
Introduction: Recommendations are mapped to Performance (25%), Innovation (25%), Operational Impact (25%), and Responsible AI (25%).
Materials and Methods: Data quality, distribution spread, quartiles, trend direction, focus-area tags, and confidence values were evaluated per metric.
Results:
- footfall: mean=306.3814, std=1082.6067, range=0.0000..7300.0000, IQR=109.0000, CV=3.534, trend=down, focus=18 anomalies, high variance, confidence=0.98.
- tempMode: mean=3.7278, std=2.6772, range=0.0000..7.0000, IQR=6.0000, CV=0.718, trend=up, focus=high variance, confidence=1.0.
- AQ: mean=4.3252, std=1.4384, range=1.0000..7.0000, IQR=3.0000, CV=0.333, trend=down, focus=high variance,

confidence=1.0.
- USS: mean=2.9396, std=1.3837, range=1.0000..7.0000, IQR=2.0000, CV=0.471, trend=up, focus=high variance, confidence=1.0.
- CS: mean=5.3941, std=1.2693, range=1.0000..7.0000, IQR=1.0000, CV=0.235, trend=down, focus=13 anomalies, high variance, confidence=0.99.
- VOC: mean=2.8422, std=2.2733, range=0.0000..6.0000, IQR=4.0000, CV=0.800, trend=up, focus=high variance, confidence=1.0.
- RP: mean=47.0434, std=16.4231, range=19.0000..91.0000, IQR=24.0000, CV=0.349, trend=up, focus=high variance, confidence=1.0.
- IP: mean=4.5657, std=1.5993, range=1.0000..7.0000, IQR=3.0000, CV=0.350, trend=down, focus=high variance, confidence=1.0.
- Temperature: mean=16.3316, std=5.9748, range=1.0000..24.0000, IQR=7.0000, CV=0.366, trend=up, focus=high variance, confidence=1.0.
- fail: mean=0.4163, std=0.4932, range=0.0000..1.0000, IQR=1.0000, CV=1.185, trend=stable, focus=high variance, confidence=1.0.

Discussion:
- Risk stratification: high risk=9, medium risk=1, low-confidence=0 metrics.
- High-priority metrics: footfall, tempMode, AQ, USS, CS, VOC, RP, IP, Temperature.
- Medium-priority metrics: fail.
- Recommendation confidence improves when calibration records, sampling context, and anomaly root-cause notes are attached to each run.

Conclusion: The dataset supports decision guidance when recommendations are executed with metric-level evidence, uncertainty disclosure, and safety checks.

Recommended Next Actions (criteria-aligned):
1) [Performance 25%] footfall: anomaly concentration detected; verify raw acquisition logs, sensor drift, and boundary thresholds.
2) [Innovation 25%] footfall: high variability (CV=3.534); add adaptive control limits and confidence-banded recommendations.
3) [Performance 25%] footfall: downward trend detected; verify if decline matches expected process behavior and acceptable bounds.
4) [Innovation 25%] tempMode: high variability (CV=0.718); add adaptive control limits and confidence-banded recommendations.
5) [Performance 25%] tempMode: upward trend detected; review upper tolerance limits and likely drift drivers.
6) [Innovation 25%] AQ: high variability (CV=0.333); add adaptive control limits and confidence-banded recommendations.
7) [Performance 25%] AQ: downward trend detected; verify if decline matches expected process behavior and acceptable bounds.
8) [Innovation 25%] USS: high variability (CV=0.471); add adaptive control limits and confidence-banded recommendations.
9) [Performance 25%] USS: upward trend detected; review upper tolerance limits and likely drift drivers.
10) [Performance 25%] CS: anomaly concentration detected; verify raw acquisition logs, sensor drift, and boundary thresholds.
11) [Operational Impact 25%] Log metric-level evidence, confidence tags, and safety outcomes in a consistent audit trail for transparent reporting.

Recommendations

Priority 1 (High Priority): Investigate sudden spikes immediately.
Priority 2 (Moderate Attention): Monitor variability weekly.
Priority 3 (Low Concern): Continue routine checks for stable metrics.

Priority Recommendation Guide

Priority Level	Meaning	Recommended Action
High Priority	Immediate attention required	Priority 1 (High Priority): Investigate sudden spikes immediately.
Moderate Attention	Track closely and follow up	Priority 2 (Moderate Attention): Monitor variability weekly.
Low Concern	Routine monitoring is sufficient	Priority 3 (Low Concern): Continue routine checks for stable metrics.

Confidence & Reliability

The analysis is generally reliable when interpreted with the same context shown in the website dashboard.
Attention is recommended for: footfall, tempMode, AQ, USS, CS, VOC, RP, IP, Temperature, fail due to unusual patterns or data quality concerns.
Missing values or unusual spikes can reduce certainty, so regular monitoring and data quality checks are advised.

Document Notes

Document Context:
- Source file: medical_patient_data.csv
- File type: csv
- Structural profile: rows=944, columns=10
- Extracted keywords: patient, diagnosis, clinical
- This report is generated for a general lab-review workflow and operational decision support.

Legal & Compliance Context

The report automatically adapts reference guidance to detected data context. Domain tags: healthcare.

• Apply recommendations with human oversight and retain an auditable decision log linking actions to evidence.

• Confirm data minimization, access control, and retention limits before operational use.

• Healthcare-related signals detected: validate protected health information controls, consent basis, and role-based access before use.

• For clinical-support use cases, require qualified human review and document intended use limitations.

AI Responsibility & Transparency

This report is generated using AI technologies provided through Microsoft Azure. Recommendations are based on available data and are intended to support decision-making. Validate critical decisions with domain experts.

Detailed Data Table

Metric	Count	Mean	Std	Min	Max
footfall	944.0	306.3814	1082.6067	0.0	7300.0

Metric	Count	Mean	Std	Min	Max
tempMode	944.0	3.7278	2.6772	0.0	7.0
AQ	944.0	4.3252	1.4384	1.0	7.0
USS	944.0	2.9396	1.3837	1.0	7.0
CS	944.0	5.3941	1.2693	1.0	7.0
VOC	944.0	2.8422	2.2733	0.0	6.0
RP	944.0	47.0434	16.4231	19.0	91.0
IP	944.0	4.5657	1.5993	1.0	7.0
Temperature	944.0	16.3316	5.9748	1.0	24.0
fail	944.0	0.4163	0.4932	0.0	1.0

References (APA 7th Edition)

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